



## STATISTICAL MECHANICS

## Lee–Yang Theorem

## Foundations, Proof and Applications



January 21, 2025

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## Abstract

In this thesis, we present the theoretical underpinnings from complex analysis for the Lee–Yang theorem and demonstrate that, under suitable conditions, the Ising model partition function remains non-vanishing within certain complex domains. The property guarantees an analytic continuation of the infinite-volume pressure and ensures the analyticity of connected correlation functions.

## 1 Introduction

The *Lee–Yang theorem* occupies a central position in statistical mechanics, offering profound insight into the analytic structure of thermodynamic functions. Originating in the seminal work of C. N. Yang and T. D. Lee in the 1950s, the theorem addresses the distribution of zeroes of the Ising-model partition function in the complex plane. One of its most celebrated manifestations is the so-called *Lee–Yang circle theorem*, which asserts that, under certain conditions, all zeroes of the finite-volume Ising partition function lie on the unit circle in the transformed complex plane. This remarkable geometric constraint has far-reaching consequences for the analyticity and phase structure of the model.

From a wider perspective, the *Lee–Yang theorem* is not confined to the Ising model alone. Its essential principle—control over the location of partition-function zeroes—generalises to various



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ferromagnetic systems. The key requirement is that the underlying statistical weights do not vanish or introduce singularities in certain complex domains, typically half-planes where the real part of the external field remains non-zero. The resulting guarantee of zero-free regions in the complex plane makes possible a holomorphic continuation of the pressure and other thermodynamic observables. In turn, this analyticity underlies many fundamental results in statistical physics, such as the characterisation of phase transitions and the exponential decay of correlations at high temperatures.

In this thesis, we first establish the necessary technical background by reviewing selected results from complex analysis: holomorphic (analytic) functions, *Cauchy's integral theorem*, the existence of antiderivatives in simply connected domains, and classical convergence theorems. We then present a systematic exposition of the *Lee–Yang theorem* itself, highlighting its precise statement, the zero-free condition on the partition function, and the consequent analyticity of the infinite-volume pressure. Further, we elucidate how this theorem extends beyond its original application in the Ising model, exploring the associated circle theorem.

Finally, we apply these theoretical developments to illustrate the analyticity of connected correlation functions, a property of fundamental importance in statistical physics. By controlling the location of partition-function zeroes and hence establishing a holomorphic extension of the pressure, it becomes possible to show that higher-order spin correlations also depend analytically on complex fields, at least within specified half-planes. These insights furnish powerful methods to investigate critical phenomena, phase transitions, and universality classes in a mathematically rigorous setting.

## 2 Theoretical Foundations in Complex Analysis

In this section, we establish the necessary foundation for understanding and proving the *Lee–Yang theorem*. We begin by introducing several essential concepts from complex analysis that will be used extensively in the arguments that follow.

**Definition 2.1** (Holomorphic Functions, [1]). Let  $D \subset \mathbb{C}$  be a domain.<sup>1</sup> A function  $f : D \rightarrow \mathbb{C}$  is said to be *holomorphic* if the derivative

$$f'(z) := \lim_{w \rightarrow z} \frac{f(w) - f(z)}{w - z}$$

exists and is finite for every  $z \in D$ .

A crucial property of holomorphic functions is that they possess derivatives of all orders, rendering them infinitely differentiable. It is noteworthy that holomorphic functions are synonymous with *analytic* functions in this thesis; that is, a function is holomorphic if and only if it can be expressed locally around any point  $z_0 \in D$  as a convergent Taylor series:

$$f(z) = \sum_{n \geq 0} a_n (z - z_0)^n,$$

where the coefficients are given by

$$a_n = \frac{1}{n!} f^{(n)}(z_0),$$

and  $z$  lies within a sufficiently small disc centred at  $z_0$ .

*Cauchy's integral theorem* provides a critical link between the path independence of integrals in the complex plane and the holomorphic property of functions.

**Theorem 2.2** (Cauchy's Integral Theorem, [1]). Let  $D \subset \mathbb{C}$  be an open, simply connected domain,<sup>2</sup> and let  $f$  be a holomorphic function defined on  $D$ . Then, for any piecewise smooth closed path  $\gamma \subset D$ , we have

$$\oint_{\gamma} f(z) dz = 0.$$

<sup>1</sup>In the context of complex analysis, a *domain* is defined as an open and connected subset of the complex plane.

<sup>2</sup>Simply connected means there are no “holes” in the domain, which would otherwise allow the possibility of non-zero integrals around closed loops.

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*Proof.* We may assume  $\gamma$  is a positively oriented simple closed contour in  $D$ , traversed so that the region  $\Omega$  it encloses lies to its left. Recall that  $\gamma$  is parametrised by  $\gamma(t) = x(t) + iy(t)$  for  $t \in [a, b]$ , where  $x(t)$  and  $y(t)$  are real-valued functions describing the  $x$ - and  $y$ -coordinates of  $\gamma$ . Writing  $f$  in terms of its real and imaginary parts

$$f(z) = u(x, y) + iv(x, y),$$

where  $u(x, y)$  and  $v(x, y)$  are real-valued functions, we can express the contour integral along  $\gamma$  as:

$$\begin{aligned} \oint_{\gamma} f(z) dz &= \int_a^b f(\gamma(t)) \gamma'(t) dt \\ &= \int_a^b [(u(x, y)x'(t) - v(x, y)y'(t)) + i(u(x, y)y'(t) + v(x, y)x'(t))] dt \\ &= \int_{\gamma} (u dx - v dy) + i \int_{\gamma} (v dx + u dy) \\ &= \iint_{\Omega} (-v_x - u_y) dx dy + i \iint_{\Omega} (u_x - v_y) dx dy && \text{(Green's theorem)} \\ &= 0. && \text{(Cauchy-Riemann equations)} \end{aligned}$$

□

**Theorem 2.3.** *Let  $f(z)$  be a holomorphic function defined on a convex open subset  $D \subset \mathbb{C}$ . Then,  $f$  has an antiderivative, i.e., there exists a holomorphic function  $F(z)$  on  $D$  such that  $F'(z) = f(z)$ .*

*Proof.* Fix a point  $z_0 \in D$ . For each  $z \in D$ , define

$$F(z) := \int_{\gamma_{z_0, z}} f(\xi) d\xi,$$

where  $\gamma_{z_0, z}(t) = (1-t)z_0 + tz$  for  $t \in [0, 1]$  is the straight-line segment joining  $z_0$  and  $z$ . Since  $f$  is holomorphic and  $D$  is convex (implying any two points in  $D$  can be joined by a straight-line segment remaining entirely within  $D$ ),  $F(z)$  is well-defined.

We aim to show that  $F'(z) = f(z)$ . For any  $w, z \in D$ , consider:

$$\begin{aligned} \frac{F(w) - F(z)}{w - z} &= \frac{1}{w - z} \left( \int_{\gamma_{z_0, w}} f(\xi) d\xi - \int_{\gamma_{z_0, z}} f(\xi) d\xi \right) \\ &= \frac{1}{w - z} \left( \int_{\gamma_{z_0, w}} f(\xi) d\xi + \int_{\gamma_{w, z}} f(\xi) d\xi + \int_{\gamma_{z, z_0}} f(\xi) d\xi \right) - \frac{1}{w - z} \int_{\gamma_{w, z}} f(\xi) d\xi \\ &= \frac{1}{w - z} \int_C f(\xi) d\xi - \frac{1}{w - z} \int_{\gamma_{w, z}} f(\xi) d\xi, \end{aligned}$$

where the function  $f$  is holomorphic on the triangle enclosed by the contour  $C = \gamma_{z_0, w} + \gamma_{w, z} + \gamma_{z, z_0}$ . By *Theorem 2.2*, we have

$$\int_C f(\xi) d\xi = 0.$$

Parametrising  $\gamma_{w, z}(t) = (1-t)w + tz$ ,  $t \in [0, 1]$ , let  $\xi = \gamma_{w, z}(t)$  so that  $d\xi = (z - w)dt$ . Substituting into the integral yields:

$$\frac{F(w) - F(z)}{w - z} = -\frac{1}{w - z} \int_0^1 f((1-t)w + tz)(z - w) dt = \int_0^1 f((1-t)w + tz) dt.$$

As  $w \rightarrow z$ , the argument of  $f$ ,  $(1-t)w + tz$ , approaches  $z$ . Since  $f$  is continuous on  $D$ , we can interchange the limit and the integral:

$$\lim_{w \rightarrow z} \frac{F(w) - F(z)}{w - z} = \int_0^1 \lim_{w \rightarrow z} f((1-t)w + tz) dt = \int_0^1 f(z) dt = f(z).$$

□

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*Remark.* (i) The theorem provides a sufficient condition for the existence of antiderivatives, but not a necessary condition. For a counterexample, consider  $(\log z)' = \frac{1}{z}$  on  $\mathbb{C} \setminus \{z \in \mathbb{C} \mid z \in \mathbb{R}, z \leq 0\}$ , which is not convex.

(ii) Any two antiderivatives of  $f$  on a connected open set differ by a constant.

**Lemma 2.4.** Every convex region is simply connected.

*Proof.* Let  $A \subset \mathbb{R}^n$  be a convex region. By definition of convexity, for any two points  $x, y \in A$ , the line segment joining  $x$  and  $y$ :  $\gamma(t) := tx + (1-t)y$ ,  $t \in [0, 1]$  lies entirely in  $A$ , implying that  $A$  is path-connected. Let  $\sigma : [0, 1] \rightarrow A$  be a loop in  $A$ , where  $\sigma(0) = \sigma(1) = p$  for some base point  $p \in A$ . We define a homotopy  $H : [0, 1] \times [0, 1] \rightarrow A$  by

$$H(s, t) = sp + (1-s)\sigma(t), \quad s, t \in [0, 1].$$

Then we verify:

- (i) *Continuity:* Since  $H(s, t)$  is a linear combination of continuous functions  $\sigma(t)$  and  $p$ ,  $H$  is continuous.
- (ii) *Well-definedness:* The convexity of  $A$  ensures that  $H(s, t) \in A$  for all  $s, t \in [0, 1]$ .
- (iii) *Boundary conditions:* When  $s = 0$ ,  $H(0, t) = \sigma(t)$  (recovering the original loop), and when  $s = 1$ ,  $H(1, t) = p$ , collapsing the loop to the constant loop at  $p$ .

Therefore, the homotopy  $H$  continuously deforms  $\sigma$  into the constant loop at  $p$  without leaving  $A$ . Since  $A$  is path-connected and every loop in  $A$  is null-homotopic,  $A$  is simply connected.  $\square$

We can now generalise the result to simply connected domains.

**Corollary 2.5.** Let  $D \subset \mathbb{C}$  be open and simply connected, and let  $f$  be holomorphic on  $D$ . Then,  $f$  has an antiderivative, i.e., there exists a holomorphic function  $F(z)$  on  $D$  such that  $F'(z) = f(z)$ .

**Theorem 2.6.** Let  $f$  be a holomorphic function on a simply connected open set  $D \subset \mathbb{C}$ , which has no zeroes on  $D$ . Then, there exists a holomorphic function  $g$  on  $D$ , called a branch of the logarithm of  $f$  on  $D$ , such that  $f = e^g$ .

*Proof.* Since  $f$  is holomorphic and non-zero on  $D$ , its derivative  $f'$  is also holomorphic on  $D$ . The function  $\frac{f'}{f}$  is thus holomorphic on  $D$ . Because  $f$  has no zeroes,  $f(z) \neq 0$  for all  $z \in D$ , implying  $\frac{f'}{f}$  is well-defined. By Corollary 2.5, there exists a holomorphic function  $F$  on  $D$  such that

$$F'(z) = \frac{f'(z)}{f(z)}, \quad \forall z \in D.$$

Define the function  $h(z) := f(z)e^{-F(z)}$ . Then we compute the derivative:

$$h'(z) = f'(z)e^{-F(z)} - f(z)F'(z)e^{-F(z)} = f'(z)e^{-F(z)} - f(z)\frac{f'(z)}{f(z)}e^{-F(z)} = 0,$$

which implies  $h(z)$  is constant on  $D$ . Therefore, there exists a constant  $c \in \mathbb{C}$  such that

$$f(z)e^{-F(z)} = c, \quad \forall z \in D.$$

Let  $g(z) := F(z) + \log c$ , where  $\log c$  is a chosen branch of the complex logarithm of  $c$ . Then  $f(z) = e^{g(z)}$ , and the holomorphic function  $g(z)$  is the desired branch of the logarithm of  $f(z)$  on  $D$ .  $\square$

*Remark.* (i) Let  $g$  be a branch of the logarithm of  $f$  on  $D$ . The real part of  $g$  satisfies  $\Re g = \log |f|$ . Indeed, observe that

$$|f| = |e^g| = e^{\Re g} \cdot |e^{i\Im g}| = e^{\Re g}.$$

Taking the natural logarithm of both sides gives  $\Re g = \log |f|$ .

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(ii) *Uniqueness of branches of the logarithm: Let  $g_1$  and  $g_2$  be two branches of the logarithm of  $f$  on  $D$ . For each  $z \in D$ ,*

$$e^{g_2(z)-g_1(z)} = \frac{e^{g_2(z)}}{e^{g_1(z)}} = \frac{f(z)}{f(z)} = 1.$$

*Hence,  $g_2(z) - g_1(z) = 2ik(z)\pi$  for some  $k(z) \in \mathbb{Z}$ . Moreover, the function  $z \mapsto k(z) = \frac{g_2(z)-g_1(z)}{2i\pi}$  is continuous and integer-valued, thus it must be constant on  $D$ , because  $D$  is connected. Hence,  $k(z) \equiv k \in \mathbb{Z}$ , which implies  $g_2 = g_1 + 2ik\pi$  for some constant  $k \in \mathbb{Z}$ .*

It is a well-established fact in complex analysis that the limit of a sequence of analytic functions does not necessarily remain analytic. However, under certain additional conditions, it is possible to ensure that the limiting function retains analyticity. To address this, recall the concept of local uniform boundedness for a family of functions.

**Definition 2.7** (Locally Uniformly Bounded, [2]). A family of functions  $\mathcal{A}$  defined on a set  $D \subset \mathbb{C}$  is said to be *locally uniformly bounded* if, for every point  $z \in D$ , there exists a real number  $M$  and a neighbourhood  $\mathcal{U} \subset D$  of  $z$  such that

$$|f(w)| \leq M, \quad \forall w \in \mathcal{U}, \quad \forall f \in \mathcal{A}.$$

**Lemma 2.8** (Schwarz's Lemma, [1]). *Let  $f$  be holomorphic on the open unit disc  $D = \{z \in \mathbb{C} : |z| < 1\}$ , with  $f(0) = 0$  and  $|f(z)| \leq 1$  for all  $z \in D$ . Then  $|f(z)| \leq |z|$  for every  $z \in D$ , and  $|f'(0)| \leq 1$ .*

*Proof.* Define

$$g(z) = \begin{cases} \frac{f(z)}{z}, & z \neq 0 \\ f'(0), & z = 0 \end{cases},$$

which is well-defined and holomorphic on  $D$  because  $\lim_{z \rightarrow 0} \frac{f(z)}{z} = f'(0)$ . Fix  $0 < r < 1$ . On  $|z| = r$ , we have  $|g(z)| = |f(z)|/r \leq 1/r$ . The maximum modulus theorem therefore gives  $|g(z)| \leq 1/r$  whenever  $|z| \leq r$ . For a fixed  $z \in D$ , letting  $r \uparrow 1$  yields  $|g(z)| \leq 1$ . Hence  $|f(z)| \leq |z|$ , and evaluating at the origin gives  $|f'(0)| = |g(0)| \leq 1$ .  $\square$

**Theorem 2.9** (Vitali's Convergence Theorem, [3]). *Let  $D \subset \mathbb{C}$  be an open, connected subset and  $(f_n)_{n \geq 1}$  a sequence of analytic functions on  $D$  that are locally uniformly bounded. Suppose that  $(f_n)_{n \geq 1}$  converges pointwise to a function  $f$  on a set  $S \subset D$ , where  $S$  has a cluster point in  $D$ . Then the sequence  $(f_n)_{n \geq 1}$  converges locally uniformly to an analytic function  $f$  on  $D$ .*

*Proof.* By local uniform boundedness and *Montel's theorem*, every subsequence of  $(f_n)$  has a further subsequence that converges locally uniformly on  $D$  to a holomorphic function  $g$ . On  $S$ , this limit agrees with the given pointwise limit  $f$ . Because  $S$  has a cluster point in  $D$ , the identity theorem implies that any two such subsequential limits coincide on all of  $D$ . If the original sequence failed to converge locally uniformly to  $g$ , there would be a compact set  $K \subset D$ , an  $\varepsilon > 0$ , and a subsequence satisfying  $\sup_K |f_n - g| \geq \varepsilon$ . *Montel's theorem* would give a further subsequence converging locally uniformly to another holomorphic limit, which must equal  $g$ , a contradiction. Hence  $(f_n)$  converges locally uniformly on  $D$  to the analytic extension of  $f$ .  $\square$

**Theorem 2.10** (Rouché's Theorem, [1]). *Let  $C$  be a simple closed contour in  $\mathbb{C}$ . Suppose  $f$  and  $g$  are two functions that are analytic inside and on  $C$ , and that*

$$|f(z)| > |g(z)|, \quad \forall z \in C.$$

*Then  $f$  and  $f + g$  have the same number of zeroes inside  $C$ , counting multiplicities.*

*Proof.* Assume that the contour  $C$  is positively oriented. Note that neither  $f$  nor  $f + g$  has zeroes on  $C$ . Define

$$h(z) := 1 + \frac{g(z)}{f(z)} = \frac{f(z) + g(z)}{f(z)}.$$

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On  $C$ , the hypothesis gives  $|h(z) - 1| < 1$ , so the curve  $h(C)$  lies in the disc centred at 1 with radius 1 and has winding number zero about the origin. By the argument principle for the meromorphic function  $h$ , this winding number equals the number of its zeroes minus the number of its poles inside  $C$ . The zeroes of  $h$  are precisely the zeroes of  $f + g$ , and its poles arise from the zeroes of  $f$ , with multiplicities counted in both cases. Their numbers are therefore equal.  $\square$

**Theorem 2.11** (Hurwitz's Theorem, [4]). *Let  $D \subset \mathbb{C}$  be an open, connected set and suppose that  $(f_n)_{n \geq 1}$  is a sequence of analytic functions on  $D$  that converges locally uniformly to an analytic function  $f$  on  $D$ . Assume that  $f_n(z) \neq 0$  for all  $z \in D$  and all  $n \geq 1$ . Then  $f$  is either identically zero or  $f(z) \neq 0$  for all  $z \in D$ .*

*Proof.* Suppose that  $f$  is not identically zero but has a zero at  $z_0 \in D$ . Choose  $R > 0$  so that  $B(z_0, R) \subset D$  and  $f$  has no zeroes on its boundary. Then

$$\delta := \min_{|z - z_0| = R} |f(z)| > 0.$$

Local uniform convergence gives  $|f_n(z) - f(z)| < \delta$  on  $|z - z_0| = R$  for all sufficiently large  $n$ . By Theorem 2.10,  $f_n$  and  $f$  then have the same number of zeroes inside the circle. Since  $f$  has at least one zero there while every  $f_n$  is zero-free, this is a contradiction. Thus a non-zero limit  $f$  has no zeroes in  $D$ .  $\square$

**Theorem 2.12** (Weierstrass' Theorem, [5]). *Let  $D \subset \mathbb{C}$  be a domain. For each  $k \in \mathbb{N}$ , let  $f_k : D \rightarrow \mathbb{C}$  be an analytic function. Suppose that the series*

$$f(z) := \sum_{k=0}^{\infty} f_k(z)$$

*is uniformly convergent on every compact subset  $K \subset D$ . Then  $f(z)$  is an analytic function on  $D$ . Moreover, for each  $n \in \mathbb{N}$ , the series of derivatives  $\sum_{k=0}^{\infty} f_k^{(n)}(z)$  converges uniformly on every compact subset  $K \subset D$ , and*

$$f^{(n)}(z) = \sum_{k=0}^{\infty} f_k^{(n)}(z), \quad \forall z \in D.$$

*Remark.* For a complete proof, see [5].

With these foundational results from complex analysis established, we are now equipped to proceed to the central argument: the *Lee–Yang theorem* in the next section.

### 3 Lee–Yang Theorem with Proof

Phase transitions are typically signalled by non-analyticities of thermodynamic quantities. Because the pressure is proportional to the logarithm of the partition function, zeroes of the complexified finite-volume partition function indicate where singular behaviour may emerge in the thermodynamic limit.

**Theorem 3.1.** *In the thermodynamic limit, the pressure*

$$\psi(\beta, h) := \lim_{\Lambda \uparrow \mathbb{Z}^d} \psi_{\Lambda}^{\#}(\beta, h)$$

*is well defined and independent of the sequence  $\Lambda \uparrow \mathbb{Z}^d$  and of the type of boundary condition. Moreover,  $\psi$  is convex (as a function on  $\mathbb{R}^* \times \mathbb{R}$ ) and is even as a function of  $h$ .*

*Remark.* For a complete proof, see [2].

We now arrive at the classical *Lee–Yang theorem*, which crucially exploits the absence of zeroes in the finite-volume partition function over certain complex domains.

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**Theorem 3.2** (Lee–Yang Theorem, [2]). *Let  $\beta \geq 0$ . Let  $D \subset \mathbb{C}$  be an open, simply connected set such that  $D \cap \mathbb{R}$  is an interval of  $\mathbb{R}$ . Suppose that, for every finite volume  $\Lambda \Subset \mathbb{Z}^d$ , the finite-volume partition function with free boundary condition (FBC) satisfies*

$$\mathbf{Z}_{\Lambda;\beta,h}^{\varnothing} \neq 0, \quad \forall h \in D. \quad (1)$$

*Then the (infinite-volume) Ising-model pressure  $\psi(\cdot)$  admits an analytic continuation to the domain  $D$ .*

In other words, although  $\psi(h)$  might initially be defined only for real  $h$  (and, *a priori*, only within a region where  $\mathbf{Z}_{\Lambda;\beta,h}^{\varnothing} \neq 0$ ), the non-vanishing condition (1) guarantees that  $\psi$  extends to a holomorphic function on the whole simply connected set  $D$ .

*Proof.* (i) *Branch of the logarithm via zero-free condition.* By hypothesis,  $\mathbf{Z}_{\Lambda;\beta,h}^{\varnothing} \neq 0$  for all  $h \in D$  and every finite  $\Lambda$ . Since  $D$  is simply connected and  $\mathbf{Z}_{\Lambda;\beta,h}^{\varnothing}$  has no zeroes there, we apply *Theorem 2.6* to show existence of a branch of the logarithm on a simply connected, zero-free domain. Thus,

$$f(h) = \mathbf{Z}_{\Lambda;\beta,h}^{\varnothing} \implies f(h) \neq 0 \text{ on } D,$$

so there exists an analytic function  $\log \mathbf{Z}_{\Lambda;\beta,h}^{\varnothing}$  defined on the whole of  $D$ , chosen so that it coincides with the usual real logarithm whenever  $h \in D \cap \mathbb{R}$ .

(ii) *Infinite-volume limit and  $g_n$ .* Let  $\Lambda_n \uparrow \mathbb{Z}^d$  be an increasing sequence of finite regions that exhaust  $\mathbb{Z}^d$  in the sense of van Hove. For each  $n$ , define

$$g_n(h) := \exp\left(\frac{1}{|\Lambda_n|} \log \mathbf{Z}_{\Lambda_n;\beta,h}^{\varnothing}\right).$$

By construction,  $g_n$  is analytic on  $D$ , because it is the exponential of an analytic function. Furthermore, for every real  $h \in D \cap \mathbb{R}$ ,  $g_n(h) = \exp(\psi_{\Lambda_n}^{\varnothing}(h))$ , where  $\psi_{\Lambda_n}^{\varnothing}(h)$  is the finite-volume pressure with FBC. By *Theorem 3.1*, the infinite-volume pressure

$$\psi(\beta, h) = \lim_{\Lambda \uparrow \mathbb{Z}^d} \psi_{\Lambda}^{\#}(\beta, h)$$

exists for all real  $h \neq 0$  and is independent of both the boundary condition and the sequence  $\Lambda \uparrow \mathbb{Z}^d$ . In particular, for the FBC,  $(\psi_{\Lambda_n}^{\varnothing}(h))_{n \geq 1}$  converges to  $\psi(\beta, h)$  for each fixed real  $h$ , i.e.,

$$g_n(h) = \exp(\psi_{\Lambda_n}^{\varnothing}(h)) \xrightarrow{n \rightarrow \infty} \exp(\psi(\beta, h)) =: g(h).$$

The function  $g : D \cap \mathbb{R} \rightarrow \mathbb{C}$  is well-defined on the real axis within  $D$ .

(iii) We now show  $\{g_n\}$  is locally uniformly bounded on  $D$ . The finite-volume partition function with FBC is

$$\mathbf{Z}_{\Lambda_n;\beta,h}^{\varnothing} = \sum_{\omega \in \Omega_{\Lambda_n}} \exp\left(-\mathcal{H}_{\Lambda_n;\beta,h}^{\varnothing}(\omega)\right).$$

For  $h \in D$ , we have

$$\begin{aligned} |\mathbf{Z}_{\Lambda_n;\beta,h}^{\varnothing}| &\leq \sum_{\omega \in \Omega_{\Lambda_n}} \left| \exp\left(-\mathcal{H}_{\Lambda_n;\beta,h}^{\varnothing}(\omega)\right) \right| \\ &= \sum_{\omega \in \Omega_{\Lambda_n}} \exp\left(-\Re \mathcal{H}_{\Lambda_n;\beta,h}^{\varnothing}(\omega)\right) \leq \exp((2d\beta + |\Re h| + \log 2)|\Lambda_n|). \end{aligned}$$

Dividing by  $|\Lambda_n|$  in the exponent and then exponentiating again for  $g_n(h)$ , we obtain

$$|g_n(h)| = \exp\left(|\Lambda_n|^{-1} \log |\mathbf{Z}_{\Lambda_n;\beta,h}^{\varnothing}|\right) \leq \exp(2d\beta + |\Re h| + \log 2).$$

Since this bound depends only on the real part of  $h$  (and not on  $n$ ), it follows that  $\{g_n\}$  is locally uniformly bounded in  $D$ .

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(iv) By *Theorem 2.9*,  $\{g_n\}$  converges locally uniformly on the entire domain  $D$  to some analytic limit  $g$ . Thus  $g : D \rightarrow \mathbb{C}$  is holomorphic and  $g_n \rightarrow g$  uniformly on compact subsets of  $D$ . Notice  $g_n(h) \neq 0$  for all  $h \in D$ ,  $n \geq 1$ . By *Theorem 2.11*, the limit  $g$  also has no zeroes in  $D$ . The possibility  $g \equiv 0$  is ruled out by continuity on the real axis: we know  $g(h) = \exp(\psi(h)) > 0$  for  $h \in D \cap \mathbb{R}$ . Hence  $g$  is nowhere zero on  $D$ .

Since  $g$  is analytic and non-zero on a simply connected domain  $D$ , we again invoke *Theorem 2.6* to obtain an analytic logarithm on  $D$ . Choose its branch so that  $\log g(h) = \psi(h)$  on  $D \cap \mathbb{R}$ . We then define  $\psi(h) := \log g(h)$  on  $D$ ; this is the required analytic continuation of the Ising-model pressure.

□

We next turn to an important corollary known as the *Lee–Yang circle theorem*. We further pinpoint the location of zeroes in the transformed plane, illustrating how they must lie on a specific geometric locus rather than in the interior of certain regions.

**Corollary 3.3** (Lee–Yang Circle Theorem, [2]). *Let  $\beta \geq 0$ . Consider the open half-planes*

$$H^+ = \{h \in \mathbb{C} : \Re h > 0\}, \quad H^- = \{h \in \mathbb{C} : \Re h < 0\}.$$

*Then, for any finite volume  $\Lambda \Subset \mathbb{Z}^d$  and FBC  $\emptyset$ , the finite-volume partition function  $\mathbf{Z}_{\Lambda;\beta,h}^\emptyset$  does not vanish anywhere in  $H^+$  or  $H^-$ . Equivalently, (1) is satisfied when  $D = H^+$  or  $D = H^-$ . Consequently, the pressure admits an analytic continuation.*

After reparametrising by  $z = e^{-2h}$ , all zeroes of the partition function lie on the unit circle  $\{z \in \mathbb{C} : |z| = 1\}$  rather than in the open unit disc  $\mathcal{U} = \{z : |z| < 1\}$ , explaining the term “circle theorem”. For a complete proof, see [2].

## 4 Applications of Lee–Yang Theorem

The *Lee–Yang theorem* finds its most direct application in the Ising model, an iconic system in statistical mechanics. In this model, one places spins  $\sigma_x \in \{-1, 1\}$  on each site  $x$  of a finite lattice  $\Lambda \subset \mathbb{Z}^d$ .<sup>3</sup> The pairwise interaction between these spins is described by the Hamiltonian

$$\mathcal{H}_\Lambda(\sigma_\Lambda) = - \sum_{\{x,y\} \subset \Lambda} J_{xy} \sigma_x \sigma_y - \sum_{x \in \Lambda} h \sigma_x, \quad (2)$$

where the couplings  $J_{xy}$  are pairwise ferromagnetic (i.e., non-negative) interactions, and  $h \in \mathbb{C}$  is an external magnetic field with  $\Re h \neq 0$ . The partition function is given by

$$\mathbf{Z}_{\Lambda,\beta}(h) = \sum_{\sigma \in \{-1,1\}^\Lambda} \exp(-\beta \mathcal{H}_\Lambda(\sigma_\Lambda)).$$

**Theorem 4.1.** *If the ferromagnetic couplings in (2) satisfy*

$$J_{xy} = J_{yx} = J_{x-y,0} \geq 0, \quad \forall x, y,$$

*and the a priori measure  $\mu_0$  on  $\mathbb{R}$  satisfies:*

(i)  $\mu_0$  is even, i.e.,  $\mu_0(d\sigma)$  is invariant under  $\sigma \mapsto -\sigma$ .

(ii) Its Laplace transform

$$\hat{\mu}_0(h) = \int_{\mathbb{R}} e^{h\sigma} d\mu_0(\sigma) \neq 0, \quad \forall h \in \mathbb{H}_+ := \{z \in \mathbb{C} : \Re z > 0\}.$$

<sup>3</sup>In the simplest and most studied variant of the Ising model, each spin takes values  $\pm 1$ . However, more general spin distributions are also admissible, provided they fulfil suitable Lee–Yang conditions that ensure the partition function does not vanish when  $\Re h \neq 0$ .

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Then, for any choice of external fields  $\{h_x\}_{x \in \Lambda}$  with  $\Re h_x > 0$  for all  $x \in \Lambda$ , the partition function satisfies

$$\mathbf{Z}_\Lambda(\{h_x\}) := \int_{\mathbb{R}^{|\Lambda|}} \exp(-\mathcal{H}_\Lambda(\sigma_\Lambda)) \prod_{x \in \Lambda} d\mu_0(\sigma_x) \neq 0.$$

*Remark.* The condition  $\hat{\mu}_0(h) \neq 0$  for  $\Re h > 0$  ensures that, in the non-interacting case ( $J_{xy} = 0$ ), the partition function does not vanish.

A significant consequence of the *Lee–Yang theorem* in this context is that key observables, such as *connected correlation functions*, exhibit analyticity in the half-planes  $\{h \in \mathbb{C} : \Re h \neq 0\}$ . These connected correlations measure the intrinsically collective behaviour of spins by discounting trivial factorisations.

**Definition 4.2** (Connected Correlation Functions, [6]). For any finite region  $\Lambda \subset \mathbb{Z}^d$ , inverse temperature  $\beta > 0$ , and homogeneous magnetic field  $h$ , the connected correlation of  $n$  spins is defined by

$$\langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\Lambda; \beta, h}^c = \left[ \frac{\partial^n}{\partial \varepsilon_1 \dots \partial \varepsilon_n} \log \left( \left\langle \exp \left[ \sum_{i=1}^n \varepsilon_i \sigma_{x_i} \right] \right\rangle_{\Lambda; \beta, h} \right) \right]_{\varepsilon_1 = \dots = \varepsilon_n = 0},$$

where  $\langle \cdot \rangle_{\Lambda; \beta, h}$  denotes the thermal average with respect to the (finite-volume) Gibbs measure associated with  $\mathcal{H}_\Lambda$ , and the superscript  $c$  denotes the connected correlation function.

*Remark.* The connected correlation functions  $\langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle^c$  can be viewed as the higher-order cumulants of the random variables  $\sigma_{x_1}, \dots, \sigma_{x_n}$ . Formally, they arise from taking derivatives of the logarithm of the moment-generating function, thereby removing contributions from lower-order correlations. In particular, for  $n = 2$ , we recover the usual covariance-like form  $\langle \sigma_x; \sigma_y \rangle^c = \langle \sigma_x \sigma_y \rangle - \langle \sigma_x \rangle \langle \sigma_y \rangle$ .

We now state the central theorem, which leverages the Lee–Yang property (non-vanishing of the partition function for  $\Re h \neq 0$ ) to demonstrate analyticity of these correlation functions in the thermodynamic limit.

**Theorem 4.3** (Analyticity of Connected Correlations, [6]). Under the assumptions above for the Ising Hamiltonian  $\mathcal{H}_\Lambda$ , with homogeneous magnetic field  $h$  and ferromagnetic couplings satisfying

$$J_{xy} = J_{yx} = J_{x-y, 0} \geq 0, \quad \forall x \neq y, \quad \sum_{x \neq 0} J_{0x} < \infty.$$

Let  $\langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\Lambda; \beta, h}^c$  be the connected correlation function of order  $n \in \mathbb{N}$ . Then, for any collection of sites  $x_1, \dots, x_n \in \mathbb{Z}^d$  and for every  $\beta > 0$ , the thermodynamic limit

$$\langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\beta, h}^c := \lim_{\Lambda \nearrow \mathbb{Z}^d} \langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\Lambda; \beta, h}^c$$

exists and defines a function of the external field  $h$  that is analytic on  $\{h \in \mathbb{C} : \Re h \neq 0\}$ .

*Proof.* Since  $\beta$  can be absorbed into a redefinition of  $J_{xy}$  and  $h$ , we set  $\beta = 1$  without loss of generality.

- (i) *Setting up a modified partition function.* We first define a modified partition function  $\mathbf{Z}_\Lambda(h, \varepsilon)$  that depends on additional complex variables  $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)$ . Let  $k_1, \dots, k_n \in \Lambda^*$ , where  $\Lambda^*$  is the dual lattice of  $\Lambda$ . Then

$$\mathbf{Z}_\Lambda(h, \varepsilon) := \int \exp \left\{ -\mathcal{H}_\Lambda^0(\sigma_\Lambda) + \sum_{x \in \Lambda} \left[ h + \sum_{\alpha=1}^n \varepsilon_\alpha e^{ik_\alpha x} \right] \sigma_x \right\} \prod_{x \in \Lambda} d\mu_0(\sigma_x),$$

where  $\mathcal{H}_\Lambda^0(\sigma_\Lambda)$  is the Hamiltonian when  $h = 0$ , and the *a priori* measure  $\mu_0$  is even and supported on  $\mathbb{R}$ . Choosing  $h_x := h + \sum_{\alpha=1}^n \varepsilon_\alpha e^{ik_\alpha x}$  ensures  $\Re h_x \geq \Re h - \sum_{\alpha} |\varepsilon_\alpha|$ . By *Theorem 4.1*, whenever  $\Re(h_x) > 0$ , the Gibbs partition function is non-vanishing:

$$\mathbf{Z}_\Lambda \left( h + \sum_{\alpha=1}^n \varepsilon_\alpha e^{ik_\alpha x} \right) \neq 0.$$

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Thus,  $\mathbf{Z}_\Lambda(h, \varepsilon) \neq 0$  as long as  $\Re h > \sum_{\alpha=1}^n |\varepsilon_\alpha|$ . Since  $\mu_0$  is even, without loss of generality we take  $\Re h > 0$ . Define the domain

$$D = \left\{ (h, \varepsilon) \in \mathbb{C}^{n+1} : \Re(h) > \sum_{\alpha=1}^n |\varepsilon_\alpha| \right\},$$

which is convex in  $\mathbb{C}^{n+1}$ . Since  $\mathbf{Z}_\Lambda(h, \varepsilon) \neq 0$  throughout  $D$ , we may choose a branch of the complex logarithm on this simply connected, zero-free region. Hence

$$f_\Lambda(h, \varepsilon) := \frac{1}{|\Lambda|} \log \mathbf{Z}_\Lambda(h, \varepsilon)$$

is well-defined and analytic in  $(h, \varepsilon) \in D$ . We then set

$$\mathcal{R}_\Lambda(h, \varepsilon) := \mathbf{Z}_\Lambda(h, \varepsilon)^{\frac{1}{|\Lambda|}} := \exp[f_\Lambda(h, \varepsilon)].$$

By construction,  $\mathcal{R}_\Lambda$  inherits the analyticity of  $\mathbf{Z}_\Lambda$  and remains non-vanishing on  $D$ .

- (ii) *Uniform bounds and thermodynamic limit.* A crucial part of the argument is to show that  $\{\mathcal{R}_\Lambda\}_\Lambda$  is locally uniformly bounded on  $D$ . Observe that

$$\begin{aligned} |\mathcal{R}_\Lambda(h, \varepsilon)| &= |\mathbf{Z}_\Lambda(h, \varepsilon)|^{\frac{1}{|\Lambda|}} \\ &\leq \left( \left\| \exp(-H_\Lambda^0 - H_\Lambda^{h, \varepsilon}) \right\|_\infty \right)^{\frac{1}{|\Lambda|}} \\ &\leq \exp\left( \frac{1}{|\Lambda|} \left( \|H_\Lambda^0\|_\infty + \|H_\Lambda^{h, \varepsilon}\|_\infty \right) \right), \end{aligned}$$

and

$$\|H_\Lambda^{h, \varepsilon}\|_\infty \leq c|\Lambda| \cdot \left( |h| + \sum_{\alpha=1}^n |\varepsilon_\alpha| \right)$$

for some constant  $c > 0$ , given that  $\mu_0$  has compact support. Likewise, using standard estimates on finite-range pair interactions  $\Phi$ , one finds

$$\|H_\Lambda^0\|_\infty \leq c'|\Lambda|,$$

where  $c'$  depends on  $\|\Phi\|$  and the lattice geometry. Thus there is a uniform exponential bound, implying

$$|\mathcal{R}_\Lambda(h, \varepsilon)| \leq \exp\left( c'' \left( |h| + \sum_{\alpha} |\varepsilon_\alpha| \right) \right)$$

for some  $c'' > 0$  independent of  $\Lambda$ . Consequently, for any compact  $K \subset D$ ,

$$\sup_{\Lambda; (h, \varepsilon) \in K} |\mathcal{R}_\Lambda(h, \varepsilon)| < \infty,$$

establishing local uniform boundedness on  $D$ .

In order to pass to the thermodynamic limit, one typically uses a cluster expansion to show that  $f_\Lambda(h, \varepsilon)$  converges pointwise to a limiting free-energy density  $f_\infty(h, \varepsilon)$ . In particular, there is a large-field regime

$$S = \{(h, \varepsilon) \in D : \Re(h) > h_0, |\Im h| < \delta, |\varepsilon_\alpha| < \delta, \forall \alpha\} \subset D,$$

on which

$$f_\infty = \lim_{\Lambda \nearrow \mathbb{Z}^d} f_\Lambda(h, \varepsilon), \quad \mathcal{R}_\infty(h, \varepsilon) = \lim_{\Lambda \nearrow \mathbb{Z}^d} \mathcal{R}_\Lambda(h, \varepsilon) = \exp(f_\infty(h, \varepsilon))$$

exist and are finite. Since  $\mathcal{R}_\Lambda$  converges pointwise to  $\mathcal{R}_\infty$  on  $S$ , and  $\{\mathcal{R}_\Lambda\}$  is uniformly bounded on compact subsets, by *Theorem 2.9*, we obtain  $\mathcal{R}_\Lambda \rightarrow \mathcal{R}_\infty$  uniformly on every compact  $K \subset D$ . By *Theorem 2.12*,  $\mathcal{R}_\infty$  is analytic on  $D$ .

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- (iii) *Non-vanishing of the limit.* We must show that  $\mathcal{R}_\infty$  is nowhere vanishing in  $D$ . It is straightforward to check that  $\mathcal{R}_\infty(h, 0) \neq 0$  for all real  $h > 0$ , by continuity from the finite-volume limit. Hence  $\mathcal{R}_\infty \neq 0$  in the entire right half-plane  $\Re(h) > 0$ . Let  $g_\Lambda^1(z) = \mathcal{R}_\Lambda(h, z, 0, \dots, 0)$ ; it converges to  $g_\infty^1(z) = \mathcal{R}_\infty(h, z, 0, \dots, 0)$  uniformly on the disc  $\{z \in \mathbb{C} : |z| < \Re(h)\}$ . A standard argument based on *Theorem 2.11* then demonstrates that  $g_\infty^1(z)$  can have no zero in the disc, and repeating this process in each variable  $\varepsilon_\alpha$  shows that  $\mathcal{R}_\infty(h, \varepsilon) \neq 0$  for  $\Re(h) > \sum_\alpha |\varepsilon_\alpha|$ . By single-valuedness of the logarithm on the simply connected, zero-free domain  $D$ , we thus write

$$f_\infty(h, \varepsilon) := \log \mathcal{R}_\infty(h, \varepsilon),$$

which is analytic in  $D$  and agrees with the thermodynamic limit of  $f_\Lambda$  wherever that limit was already known to exist. Since  $f_\Lambda \rightarrow f_\infty$  uniformly on compact subsets of  $D$ , the same applies to any finite set of partial derivatives, by *Theorem 2.2*.

- (iv) *Connection to correlation functions.* Finally, we link these analytically extended free-energy functions to the connected correlations. Consider the (discrete) Fourier transform

$$\hat{\sigma}_k = \sum_{x \in \Lambda} e^{ikx} \sigma_x, \quad k \in \Lambda^*,$$

and its inverse

$$\sigma_x = \frac{1}{|\Lambda|} \sum_{k \in \Lambda^*} e^{-ikx} \hat{\sigma}_k, \quad x \in \Lambda.$$

The partition function  $\mathbf{Z}_\Lambda(h, \varepsilon)$  can be rewritten as

$$\mathbf{Z}_\Lambda(h, \varepsilon) = \mathbf{Z}_\Lambda(h) \left\langle \exp \left( \sum_{\alpha=1}^n \varepsilon_\alpha \hat{\sigma}_{k_\alpha} \right) \right\rangle_{\Lambda, h}.$$

Differentiating repeatedly with respect to  $\varepsilon_\alpha$  gives

$$\left. \frac{\partial^n \log \mathbf{Z}_\Lambda(h, \varepsilon)}{\partial \varepsilon_1 \cdots \partial \varepsilon_n} \right|_{\varepsilon=0} = \left\langle \hat{\sigma}_{k_1}; \dots; \hat{\sigma}_{k_n} \right\rangle_{\Lambda, h}^c \Big|_{\sum_{\alpha=1}^n k_\alpha = [0]},$$

where the condition  $\sum_{\alpha=1}^n k_\alpha = [0]$  reflects translational invariance under the imposed PER. Hence

$$\left. \frac{\partial^n f_\Lambda(h, \varepsilon)}{\partial \varepsilon_1 \cdots \partial \varepsilon_n} \right|_{\varepsilon=0} = \frac{1}{|\Lambda|} \left\langle \hat{\sigma}_{k_1}; \dots; \hat{\sigma}_{k_n} \right\rangle_{\Lambda, h}^c \Big|_{\sum_{\alpha=1}^n k_\alpha = [0]}.$$

Taking the limit as  $\Lambda \nearrow \mathbb{Z}^d$  and Fourier-transforming back to position space, one concludes that

$$\langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\beta, h}^c := \lim_{\Lambda \nearrow \mathbb{Z}^d} \langle \sigma_{x_1}; \dots; \sigma_{x_n} \rangle_{\Lambda; \beta, h}^c$$

is an analytic function of  $h$  on  $\{\Re(h) \neq 0\}$ .

□

## 5 Conclusion

In conclusion, we have established a rigorous framework, grounded in the fundamental concepts of complex analysis, to underpin and prove the *Lee–Yang theorem*. By invoking key results such as *Cauchy’s integral theorem*, *Rouché’s theorem*, and *Hurwitz’s theorem*, we have demonstrated that the partition function of the Ising model remains non-zero across specific domains in the complex plane, thereby enabling an analytic continuation of the pressure. In particular, the absence of zeroes in half-planes with non-zero real part of the external field underpins the celebrated *Lee–Yang circle theorem*, which locates all zeroes on the unit circle under a suitable reparametrisation.

Moreover, this analytic structure facilitates profound implications for physical observables in the thermodynamic limit: connected correlation functions are shown to be analytic in the same

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half-planes, thereby elucidating the nature of critical phenomena and phase transitions within the Ising model and beyond. The synergy between complex analysis and statistical mechanics, as exemplified by the *Lee–Yang theorem*, not only provides a mathematically robust description of ferromagnetic systems, but also invites further exploration into more general classes of models exhibiting similar zero-free regions.

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